

REACT QUESTION PAPER

TIME: 1.5 HOURS

TOTAL MARKS: 50

Section A – Theory (2 Marks Each)

Answer all questions

1. What is **React.js**? Mention two advantages of using React.
2. What is the difference between **React components** and **HTML elements**?
3. What is the difference between **functional components** and **class components**?
4. What are **props** in React?
5. What is **state** in React? How is it different from props?
6. What is the purpose of the **useState()** hook?
7. What is the **Virtual DOM** and how does it improve performance?
8. What is the purpose of the **useEffect()** hook?
9. What is the difference between **controlled components** and **uncontrolled components**?
10. What are **React hooks**?
11. What is **React Router** and why is it used?
12. What is **lifting state up** in React?

13. What is the difference between **map()** and **forEach()** when rendering lists in React?
14. What are **keys in React lists** and why are they important?
15. What is **context API** in React?

Section B – Programs (5 Marks Each)

Answer all questions

1. Create a React component that takes a list of users and displays them using map().

Example Data:

```
[  
  {id:1, name:"John"},  
  {id:2, name:"Alice"},  
  {id:3, name:"David"}  
]
```

Display the names as a **list in the UI**.

2. Create a React form component with the following fields:

- Name
- Email
- Password

Use **useState()** to manage the form data and display the entered data after submission.

3. Create a React application that fetches data from an API using `useEffect()` and `fetch()`.

Example API:

<https://jsonplaceholder.typicode.com/posts>

Display post titles in a list.

4. Create a React application with React Router that contains the following pages:

- Home
- About
- Contact

Each page should display different content when the route changes.